OL Organic Letters

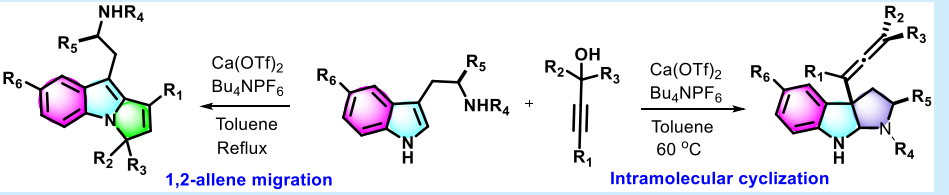
pubs.acs.org/OrgLett Letter

Ca(II)-Catalyzed Cascade Reaction of Tryptamines with Propargylic Alcohols: Temperature-Driven Ring Opening and Closing via the Allene Migration Pathway for the Synthesis of Pyrrolo[1,2-*a*]indoles

Ashok Kale, Su Jeong Kwon, Joohan Lee, Jae Kyun Lee, and Kyeong Lee*

Cite This: *Org. Lett.* 2024, 26, 1196−1200 Read Online

ACCESS | Metrics & More | Article Recommendations | Supporting Information



ABSTRACT: A temperature-dependent cascade of reactions between tryptamines and propargylic alcohols was developed to achieve selective formation of pyrroloindoline and pyrrolo[1,2-*a*]indole heterocycles by Ca(II) catalysis. The cascade consists of electrophilic addition of allene at the C3 carbon of indole followed by intramolecular cyclization at 60 °C to yield pyrroloindolines. Furthermore, simultaneous 1,2-allene migration and pyrrolidine ring opening were followed by intramolecular cyclization via C−N bond formation at reflux temperature to obtain pyrrolo[1,2-*a*]indole scaffolds. A wide range of substrates, a clean reaction profile, scalability, and good to excellent yields are the advantages of this protocol.

Downloaded via FUDAN UNIV on February 5, 2025 at 03:01:33 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

Indole and its derivatives have attained considerable importance in synthetic chemistry owing to their broad biological profile and versatile chemical reactivities.[1] Among them, the pyrroloindoline core structure is found in numerous natural products (Figure 1)[2] and is a common structural motif among secondary metabolites of terrestrial and marine origin.[3] Especially, C3 carbon-derived pyrroloindoline shows a distinctive range of chemical and biological properties. For example, *Calycanthaceae* plant extracts have been used as traditional Chinese medicines for the treatment of fungal infection,[4] melanogenesis, tumors, hypertension, and inflammatory conditions.[5] On the other hand, pyrrolo[1,2-*a*]indoles and their scaffolds are an important class of privileged structural units that are found in many natural products and

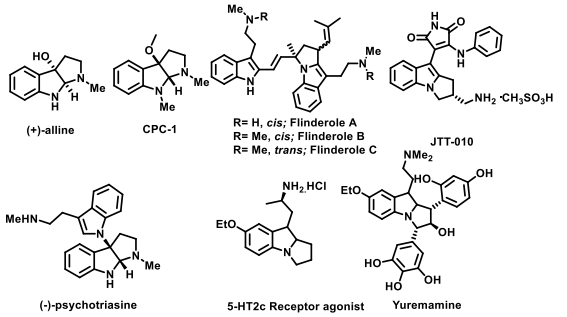
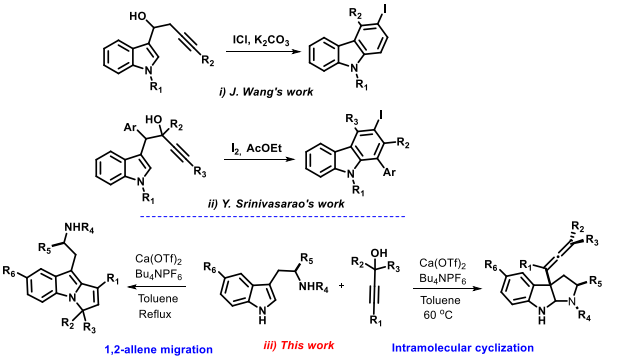
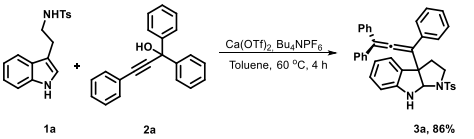


Figure 1. Natural products containing a pyrroloindoline and pyrrolo[1,2-*a*] indole unit.

Scheme 1. Migration Strategies of Indole



Scheme 2. Synthesis of Compound 3a



Received: December 27, 2023
Revised: January 23, 2024
Accepted: January 29, 2024
Published: February 2, 2024



© 2024 American Chemical Society

https://doi.org/10.1021/acs.orglett.3c04357
Org. Lett. 2024, 26, 1196−1200

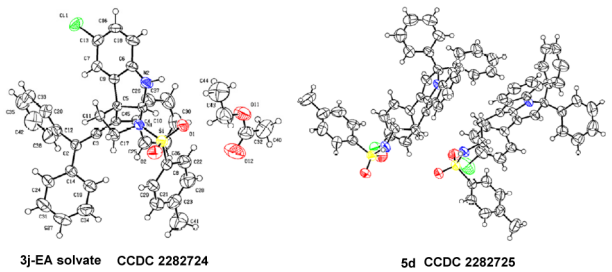
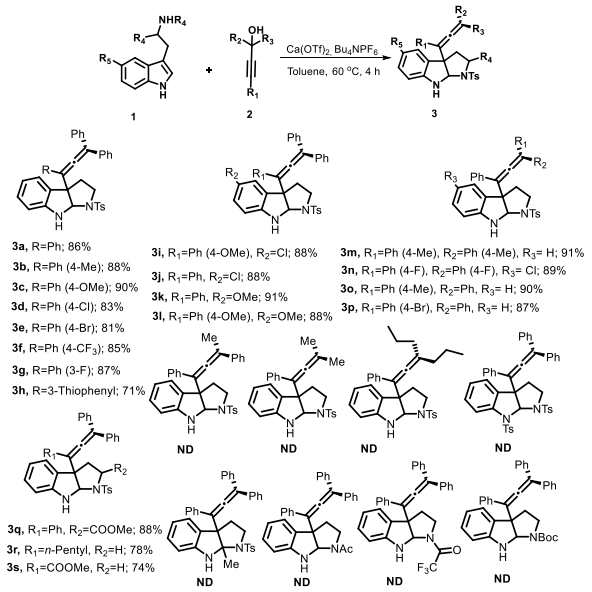


Figure 2. ORTEP diagrams of **3j** and **5d** showing 50% probability displacement ellipsoids.

pharmaceuticals that exhibit a wide range of biological activities (Figure 1).[6] Specifically, JTT-10 is a protein kinase C-β-selective inhibitor.[7] Flinderoles A, B, and C show selective antimalarial activities.[8] In addition, these are key intermediates and play a significant role in the synthesis of many complex natural products and biologically active molecules.[9] Therefore, the development of novel and efficient protocols toward the synthesis of such molecules has received considerable attention. Few research groups have studied the reactivity of indoles at the C3 position followed by alkyl migration reactions.[10] However, pyrrolo[1,2-*a*]indole synthesis from these starting materials has not been well explored. Wang et al. reported the synthesis of 3-iodocarbazoles using ICl/K_2CO_3 from α-hydroxy substrates.[11] Later, Wang et al. demonstrated divergent reactions between tryptophols and propargylic alcohols in the presence of $BF_3{\cdot}OEt_2$.[12] Recently, Yaragorla et al. reported iodo-cyclo isomerization of 1-(indol-3-yl)-1-arylbut-3-yn-2-ols to form 3-iodocarbazoles (Scheme 1).[13] As part of our ongoing research, we aimed to synthesize biologically active molecules and natural products.[14] Herein, we report for the first time a novel route for the synthesis of pyrrolo[1,2-*a*] indoles via allene migration, followed by pyrrolidine ring opening through Ca(II)-catalyzed cascade reactions of tryptamines and propargylic alcohols. The synthesis of allene-substituted pyrroloindoline began by reacting *N*-(2-(1*H*-indol-3-yl) ethyl)-4-methylbenzenesulfonamide **1a** (0.31 mmol) and propargyl alcohol **2a** (0.31 mmol) with 5/5 mol % of $Ca(OTf)_2/Bu_4NPF_6$ in toluene at room temperature for 12 h; no product formation was observed. The reaction temperature was then increased to 60 °C, which resulted in the complete disappearance of starting materials and the formation of the product observed after 4 h with 86% yield (Scheme 2). The structure of the product was confirmed by spectral data NMR, HRMS, and single-crystal X-ray diffraction (**3j**) (Figure 2).

Table 2. Substrate Scope of 3[a]



[a]General conditions: **1a** (0.31 mmol, 1.0 equiv), **2a** (0.31 mmol, 1.0 equiv), $Ca(OTf)_2$ (0.015 mmol, 0.05 equiv), and Bu_4NPF_6 (0.015 mmol, 0.05 equiv) in toluene (5 mL) at 60 °C for 4 h. [b]Isolated yields, ND = not detected.

Scheme 3. Synthesis of 5a

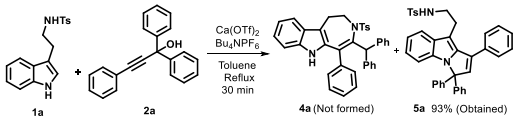


Table 1. Optimization of the Reaction Conditions for 3a[a]

Entry	Catalyst (mol %)	Solvent	Temp. (°C)	Time	Yield[b] (%), 3a	Yield[b] (%), C
1	*X*/*Y* (5/5)	Toluene	rt	12 h	NR	np
2	*X*/*Y* (5/5)	Toluene	40	12 h	32	np
3	*X*/*Y* (5/5)	Toluene	50	12 h	74	np
4	***X*/*Y* (5/5)**	**Toluene**	**60**	**4 h**	**86**	**np**
5	*X*/*Y* (5/5)	Toluene	70	4 h	78	trace
6	*X*/*Y* (5/5)	Toluene	80	4 h	72	24
7	*X*/*Y* (1/1 equiv)	Toluene	60	2 h	56	37
8	*X*/*Y* (5/5)	ACN	60	4 h	75	np
9	*X*/*Y* (5/5)	THF	60	4 h	32	np
10	*X*/*Y* (5/5)	DCE	60	4 h	76	np
11	*X*/*Y* (5/5)	Water	60	4 h	NR	np
12	*X* (10)	Toluene	60	4 h	63	np
13	$Cu(OTf)_2$ (10)	Toluene	60	4 h	55	np
14	$Sc(OTf)_2$ (10)	Toluene	60	4 h	68	np
15	$Yt(OTf)_2$ (10)	Toluene	60	4 h	71	np
16	$Zn(OTf)_2$ (10)	Toluene	60	4 h	74	np

[a]General conditions: **1a** (0.31 mmol, 1.0 equiv), **2a** (0.31 mmol, 1.0 equiv), ***X*** = $Ca(OTf)_2$ (0.015 mmol, 0.05 equiv), and ***Y*** = Bu_4NPF_6 (0.015 mmol, 0.05 equiv) in toluene (5 mL) at 60 °C for 4 h. [b]Isolated yields, np = no product, temp. = temperature, and equiv = equivalent.

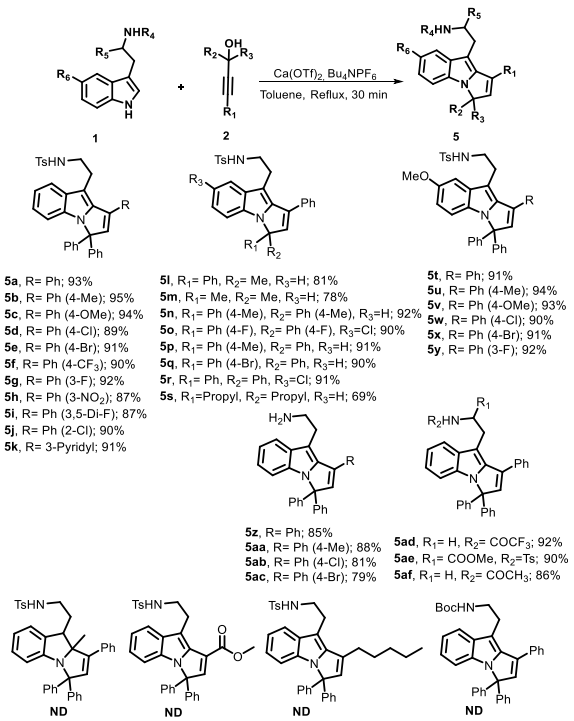
https://doi.org/10.1021/acs.orglett.3c04357
Org. Lett. 2024, 26, 1196−1200

Table 3. Optimization of the Reaction Conditions for 5a[a]

Entry	Catalyst (mol %)	Solvent	Temperature	Time	Yield[b] (%)
1	$Ca(OTf)_2$ (10)	Toluene	Reflux	30 min	75
2	***X*/*Y*** **(5/5)**	**Toluene**	**Reflux**	**30 min**	**93**
3	***X*/*Y*** (1/1 equiv)	Toluene	Reflux	30 min	93
4	$Sc(OTf)_2$ (10)	Toluene	Reflux	2 h	85
5	$Cu(OTf)_2$ (10)	Toluene	Reflux	2 h	81
6	$Yt(OTf)_2$ (10)	Toluene	Reflux	2 h	78
7	$Zn(OTf)_2$ (10)	Toluene	Reflux	2 h	80
8	*p*TSA (10)	Toluene	Reflux	2 h	31
9	CSA (10)	Toluene	Reflux	2 h	45
10	***X*/*Y*** (5/5)	ACN	Reflux	2 h	45
11	***X*/*Y*** (5/5)	THF	Reflux	2 h	np
12	***X*/*Y*** (5/5)	DCE	Reflux	2 h	75
13	***X*/*Y*** (5/5)	Water	Reflux	2 h	np

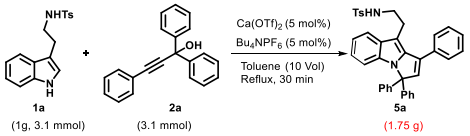
[a]General conditions: **1a** (0.31 mmol, 1.0 equiv), **2a** (0.31 mmol, 1.0 equiv), ***X*** = $Ca(OTf)_2$ (0.015 mmol, 0.05 equiv), and ***Y*** = Bu_4NPF_6 (0.015 mmol, 0.05 equiv) in toluene (5 mL) at reflux temperature for 30 min. [b]Isolated yields, np = no product, and equiv = equivalent.

Table 4. Substrate Scope of 5[a]



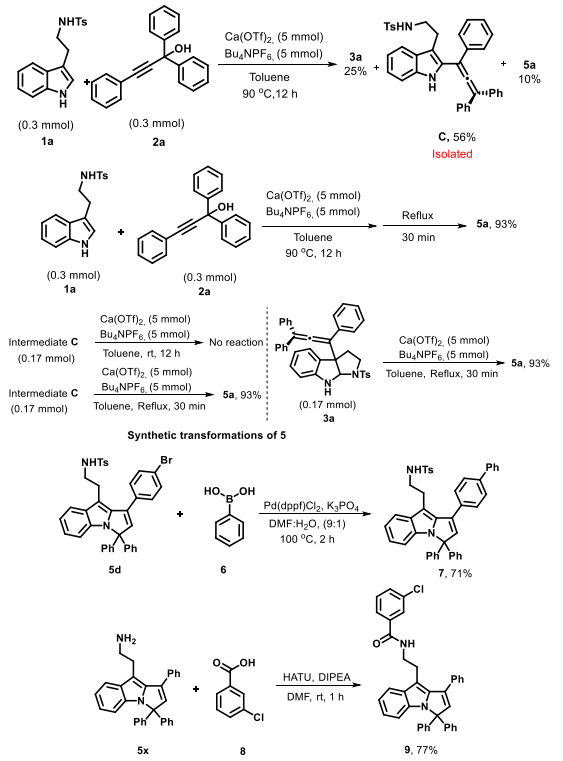
[a]General conditions: **1a** (0.31 mmol, 1.0 equiv), **2a** (0.31 mmol, 1.0 equiv), $Ca(OTf)_2$ (0.015 mmol, 0.05 equiv), and Bu_4NPF_6 (0.015 mmol, 0.05 equiv) in toluene (5 mL) at reflux temperature for 30 min. [b]Isolated yields, ND = Not detected.

Scheme 4. Gram-Scale Synthesis of 5a

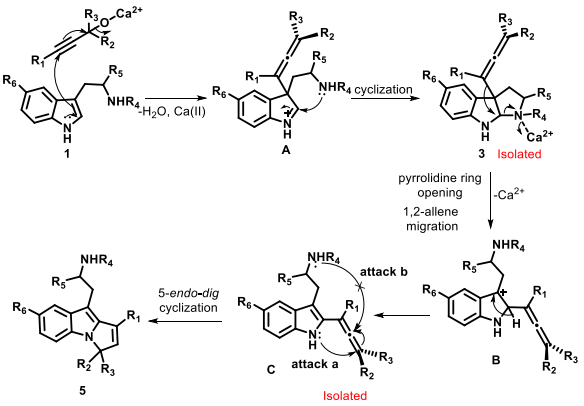


Inspired by these results, reactions were conducted to optimize the formation of **3a**. First, a series of reactions were conducted by changing the reaction temperature, which

Scheme 5. Controlled Experiments and Synthetic Transformations



Scheme 6. Plausible Mechanism



resulted in no drastic change in yield (Table 1, entries 3−6). However, in addition to compound **3a**, an intermediate **C** was observed as shown in the mechanism (Table 1, entries 5 and 6). Next, the feasibility of the reaction was assessed by changing the solvent (Table 1, entries 8−11). The results showed that toluene was the best solvent for this reaction.

Furthermore, to evaluate the role of additive (Bu_4NPF_6), the conditions of entry 4 were repeated without it, and a slow reaction rate and low yield were observed (entry 12). We conclude that the additive played a key role by increasing the Lewis acidity of Ca(II) by exchanging its counteranion.[15] Finally, a reaction was conducted with a stoichiometric ratio of reagents (Table 1, entry 7), and we found a mixture of products. Other catalysts, such as $Cu(OTf)_2$, $Sc(OTf)_2$, $Yt(OTf)_2$, and $Zn(OTf)_2$, were found to be moderately effective in catalyzing this reaction (Table 1, entries 13−16). Using the optimized reaction conditions, the scope and

https://doi.org/10.1021/acs.orglett.3c04357
Org. Lett. 2024, 26, 1196−1200

generality of this transformation were further evaluated by varying the different tryptamines like NHTs, $NHCOCF_3$, $NHCOCH_3$, indole NTs, and NHBoc (Table 2). However, except for NHTs, the tryptamines remaining failed to produce the desired product; this fact might be due to the stability of the partial pyramidalization of the sulfonamidate in the transition state.[16] Propargylic alcohols prepared from aliphatic ketones and acetophenone could not produce the corresponding products due to less reactivity under standard reaction conditions (low temperature).

After assessing the generality of the reaction with various tryptamines and propargylic alcohols (Table 2), an experiment was conducted at reflux temperature for the exclusive formation of the intermediate **C** followed by intramolecular cyclization to obtain the possible products **4a** or **5a**. Interestingly, the formation of pyrrolo[1,2-*a*]indole (compound **5a**) was observed (Scheme 3) which was confirmed by NMR, HRMS, and single-crystal X-ray data (**5d**) (Figure 2). With this result, we attempted to optimize the reaction conditions to obtain the desired compound **4a** using intermediate **C**, but no results were satisfactory. Experiments were subsequently conducted to optimize **5a** with a stoichiometric ratio of catalyst, additive, and other catalysts such as $Sc(OTf)_2$, $Cu(OTf)_2$, $Yt(OTf)_2$, $Zn(OTf)_2$, CSA, and *p*TSA. Of these, CSA and *p*TSA produced low yields, whereas $Sc(OTf)_2$, $Cu(OTf)_2$, $Yt(OTf)_2$, and $Zn(OTf)_2$ produced moderate yields (Table 3, entries 3−9). Next, we evaluated different solvents to determine their effect on the reaction yield, but no change in the yield of **5a** was observed (Table 3, entries 10−13). Using the optimized reaction conditions for the synthesis of **5a**, the substrate scope was evaluated by reacting to various substituted tryptamines NHTs, $NHCOCF_3$, $NHCOCH_3$, and NHBoc (Table 4). However, in the reaction using NHBoc tryptamine, Boc deprotected product **5z** was observed under standard conditions due to the Ca(II) reagent and high temperature. Furthermore, propargylic alcohols prepared from aliphatic alkynes failed to give the desired products. This might be due to the lack of migration efficiency of allene. These results showed that this protocol produced the desired compounds in high yields (Table 4). After the successful formation of **5a**, a gram-scale reaction was further performed to assess the practicality of this protocol during the scale-up process (Scheme 4). Therefore, we performed the reaction with 3.1 mmol of **1a** (1 g) and **2a** under standard reaction conditions (Table 3, entry 2) and isolated 1.75 g of **5a**. Furthermore, a stepwise mechanistic study was conducted for the formation of **5a**. Initially, a model reaction was carried out at 90 °C for 12 h to isolate intermediate **C**. However, a mixture of products was obtained (**3a** 25%, **C** 56%, and **5a** 10%). Then the same reaction was repeated, and after 12 h, the reaction mass was further heated to reflux temperature. Surprisingly, exclusive formation of **5a** was observed with a yield of 93%. These results showed that compound **3a** was also converted to **5a** at the reflux temperature, and this was reconfirmed by the reaction of **3a** under standard conditions (Scheme 5).

The synthetic utility of the selected compounds of **5** is demonstrated by treating **5d** with phenylboronic acid under Suzuki cross-coupling conditions to give compound **7**. In addition, conducting an acid−amine coupling reaction of **5z** with 3-chlorobenzoic acid leads to the formation of **9** (Scheme 5).

The detailed reaction mechanism for the formation of **3** and **5** is shown in Scheme 6. Initially, $Ca(OTf)_2/Bu_4NPF_6$ activated propargylic alcohol **2** undergoes an S_N2' reaction with **1** that results in iminium **A**. Intramolecular cyclization of **A** leads to the formation of **3a** via C−N bond formation of NHTs with imine carbon. Next, 1,2-allene migration followed by pyrrolidine ring opening leads to **B**, which on *β*-H elimination forms 2-allenyltryptamine **C**. Finally, intermediate **C** undergoes cyclization via a 5-*exo*-*dig* manner to produce **5**.

In conclusion, we demonstrated selective product formation at different temperatures by Ca(II)-catalyzed cascade reactions of tryptamines and propargylic alcohols via pyrrolidine ring opening by C−C bond, allene migration, and a new C−N bond formation mechanism. In addition, we evaluated the role of Bu_4NPF_6 and detailed mechanistic studies were conducted to support the formation of pyrrolo[1,2-*a*]indoles. Furthermore, the proposed protocol was scalable, and mild reaction conditions, low reaction time, easy procedures, a wide range of substrate scope, and high yields are the advantages of this protocol.

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.orglett.3c04357.

Experimental procedures, optimization tables, analytical data, and copies of the ^1H and ^{13}C NMR spectra for all new products (PDF)

Accession Codes

CCDC 2282724−2282725 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

■ AUTHOR INFORMATION

Corresponding Author

Kyeong Lee − *BK21 FOUR Team and Integrated Research Institute for Drug Development, College of Pharmacy, Dongguk University, Seoul, Goyang 10326, Republic of Korea*; orcid.org/0000-0002-5455-9956; Email: kaylee@dongguk.edu

Authors

Ashok Kale − *BK21 FOUR Team and Integrated Research Institute for Drug Development, College of Pharmacy, Dongguk University, Seoul, Goyang 10326, Republic of Korea*

Su Jeong Kwon − *BK21 FOUR Team and Integrated Research Institute for Drug Development, College of Pharmacy, Dongguk University, Seoul, Goyang 10326, Republic of Korea*

Joohan Lee − *BK21 FOUR Team and Integrated Research Institute for Drug Development, College of Pharmacy, Dongguk University, Seoul, Goyang 10326, Republic of Korea*

Jae Kyun Lee − *Brain Science Institute, KIST, Seoul 02792, Republic of Korea*; orcid.org/0000-0003-0587-5319

Complete contact information is available at:
https://pubs.acs.org/10.1021/acs.orglett.3c04357

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This study was supported by the National Research Foundation of Korea (NRF) funded by the Korean government (MSIT) [No. 2018R1A5A2023127 and No. 2023R1A2C3004599] and the BK21 FOUR program funded by the Ministry of Education of Korea through NRF.

■ REFERENCES

(1) (a) Sundberg, R. J. *The Chemistry of Indoles*; Academic Press: London, 1970. (b) Gribble, G. W. *Heterocyclic Scaffolds II: Reactions and Applications of Indoles*; Springer: Berlin, 2010.

(2) For leading reviews, see: (a) Crich, D.; Banerjee, A. Chemistry of the hexahydropyrrolo [2,3-b] indoles: Configuration, conformation, reactivity, and applications in synthesis. *Acc. Chem. Res.* **2007**, *40*, 151−161. (b) Ruiz-Sanchis, P.; Savina, S. A.; Albericio, F.; Álvarez, M. Structure, Bioactivity and synthesis of natural products with hexahydropyrrolo[2,3-b] indole. *Chem. Eur. J.* **2011**, *17*, 1388−1408. (c) Büschleb, M.; Dorich, S.; Hanessian, S.; Tao, D.; Schenthal, K. B.; Overman, L. E. Synthetic strategies toward natural products containing contiguous stereogenic quaternary carbon atoms. *Angew. Chem., Int. Ed.* **2016**, *55*, 4156−4186.

(3) Ruiz-Sanchis, P.; Savina, S. A.; Albericio, F.; Alvarez, M. Structure, bioactivity and synthesis of natural products with hexahydropyrrolo [2,3-*b*] indole. *Chem. Eur. J.* **2011**, *17*, 1388−1408.

(4) Bowman, W. R.; Cloonan, M. O.; Fletcher, A. J.; Stein, T. Synthesis of heteroarenes using cascade radical cyclisation via iminyl radicals. *Org. Biomol. Chem.* **2005**, *3*, 1460−1467.

(5) (a) Gui, R. Y.; Liang, W. W.; Yang, S. X.; Llu, L.; Qin, J. C. Chemical composition, antifungal activity and toxicity of essential oils from the leaves of Chimonanthus praecox located at two different geographical origins. *Asian J. Chem.* **2014**, *26*, 4445−4448. (b) Araki, T.; Manabe, Y.; Fujioka, K. Total syntheses of (±)-folicanthine and (±)-chimonanthine via a double intramolecular carbamoylketene−alkene [2 + 2] cycloaddition. *Tetrahedron Lett.* **2013**, *54*, 1012−1014. (c) Zhang, J. W.; Gao, J. M.; Xu, T. Antifungal activity of alkaloids from the seeds of Chimonanthus praecox. *Chem. Biodivers.* **2009**, *6*, 838−845.

(6) For a selected review, see: (a) Bass, P. D.; Gubler, D. A.; Judd, T. C.; Williams, R. M. Mitomycinoid alkaloids: Mechanism of action, biosynthesis, total syntheses, and synthetic approaches. *Chem. Rev.* **2013**, *113*, 6816. (b) Galm, U.; Hager, M. H.; Van Lanen, S. G.; Ju, J.; Thorson, J. S.; Shen, B. Antitumor antibiotics: Bleomycin, enediynes, and mitomycin. *Chem. Rev.* **2005**, *105*, 739−758.

(7) For selected examples, see: (a) Tanaka, M.; Sagawa, S.; Hoshi, J. I.; Shimoma, F.; Yasue, K.; Ubukata, M.; Ikemoto, T.; Hase, Y.; Takahashi, M.; Sasase, T.; Ueda, N.; Matsushita, M.; Inaba, T. Synthesis, SAR studies, and pharmacological evaluation of 3-anilino-4-(3-indolyl) maleimides with conformationally restricted structure as orally bioavailable PKCbeta-selective inhibitors. *Bioorg. Med. Chem.* **2006**, *14*, 5781. (b) Tanaka, M.; Ubukata, M.; Matsuo, T.; Yasue, K.; Matsumoto, K.; Kajimoto, Y.; Ogo, T.; Inaba, T. One-step synthesis of heteroaromatic-fused pyrrolidines via cyclopropane ring-opening reaction: application to the PKCbeta inhibitor JTT-010. *Org. Lett.* **2007**, *9*, 3331−3334.

(8) (a) Fernandez, L. S.; Buchanan, M. S.; Carroll, A. R.; Feng, Y. J.; Quinn, R. J.; Avery, V. M.; Flinderoles, A. C. Flinderoles A−C: Antimalarial bis-indole alkaloids from Flindersia species. *Org. Lett.* **2009**, *11*, 329−332. (b) Dethe, D. H.; Erande, R. D.; Ranjan, A. Biomimetic total syntheses of flinderoles B and C. *J. Am. Chem. Soc.* **2011**, *133*, 2864−2867. (c) Zeldin, R. M.; Toste, F. D. Synthesis of flinderoles B and C by a gold-catalyzed allene hydroarylation. *Chem. Sci.* **2011**, *2*, 1706. (d) Vallakati, R.; May, J. A. Biomimetic synthesis of the antimalarial flindersial alkaloids. *J. Am. Chem. Soc.* **2012**, *134*, 6936−6939.

(9) For selected examples, see: (a) Colandrea, V. J.; Rajaraman, S.; Jimenez, L. S. Synthesis of the mitomycin and FR900482 ring systems via dimethyldioxirane oxidation. *Org. Lett.* **2003**, *5*, 785−787. (b) Dethe, D. H.; Erande, R. D.; Ranjan, A. Biomimetic total syntheses of borreverine and flinderole alkaloids. *J. Org. Chem.* **2013**, *78*, 10106−10120.

(10) (a) Tharra, P.; Baire, B. Regioselective cyclization of (idol-3-yl)pentyn-3-ols as an approach to (tetrahydro)carbazoles. *Org. Lett.* **2018**, *20*, 1118−1121. (b) Chen, L.; Zou, Y.-X.; Fang, X.-Y.; Wu, J.; Sun, X.-H. Brønsted acid-catalyzed regio divergent phosphorylation of 2-indolylmethanols to synthesize benzylic site or C3-phosphorylate-dindole derivatives. *Org. Biomol. Chem.* **2018**, *16*, 7417−7424. (c) Cui, B.; Gao, J.; Fan, L.; Jiao, Y.; Lu, T.; Feng, J. Dehydroxylated C-3 Alkylation of Indole Accompanied by 1,2-Sulfur Migration. *J. Org. Chem.* **2020**, *85*, 6206−6215. (d) Mutra, M. R.; Chandana, T. L.; Wang, Y.-J.; Wang, J.-J. Green and rapid acid-catalyzed ynamide skeletal rearrangement and stereospecific functionalization with anisole derivatives. *Green Chem.* **2023**, *25*, 8124−8133. (e) Mutra, M. R.; Wang, J.-J. Photoinduced ynamide structural reshuffling and functionalization. *Nat. Commun.* **2022**, *13*, 2345.

(11) Wang, J.; Zhu, H.-T.; Qiu, Y.-F.; Niu, Y.; Chen, S.; Li, Y.-X.; Liu, X.-Y.; Liang, Y.-M. Facile synthesis of carbazoles via a tandem iodocyclization with 1,2-alkyl migration and aromatization. *Org. Lett.* **2015**, *17*, 3186−3189.

(12) Huang, K.; Sheng, G.; Lu, P.; Wang, Y. BF3-promoted divergent reactions between tryptophols and propargylic alcohols. *Org. Lett.* **2017**, *19*, 4114−4117.

(13) Yaragorla, S.; Bag, D.; Dada, R.; Jovan Jose, K. V. Iodo-Cycloisomerization of Aryl(indol-3-yl)methane-Tethered Propargyl Alcohols to 3-Iodocarbazoles via Selective 1,2-Alkyl Migration. *ACS Omega* **2018**, *3*, 15024−15034.

(14) (a) Nada, H.; Elkamhawy, A.; Lee, K. Structure-activity relationship of key heterocyclic anti-angiogenic leads of promising potential in the fight against cancer. *Molecules* **2021**, *26*, 553. (b) Lee, K.; Ban, H. S.; Naik, R.; Hong, Y. S.; Son, S.; Kim, B.-K.; Xia, Y.; Song, K. B.; Lee, H.-S.; Won, M. Identification of malate dehydrogenase 2 as a target protein of the HIF-1 inhibitor LW6 using chemical probes. *Angew. Chem., Int. Ed.* **2013**, *52*, 10286−10289. (c) Naik, R.; Won, M.; Kim, B. K.; Xia, Y.; Choi, H. K.; Jin, G.; Jung, Y.; Kim, H. M.; Lee, K. Synthesis and structure-activity relationship of (E)-phenoxy acrylic amide derivatives as hypoxia-inducible factor (HIF) 1α inhibitors. *J. Med. Chem.* **2012**, *55*, 10564−1057. (d) Lu, Q.; Harmalkar, D. S.; Quan, G.; Kwon, H.; Cho, J.; Choi, Y.; Lee, D.; Lee, K. Total synthesis of the neuroprotective agent cudraisoflavone J. *J. Nat. Prod.* **2021**, *84*, 1359−1365.

(15) For reviews on calcium catalysis, see: (a) Begouin, J. M.; Niggemann, M. Calcium-based Lewis acid catalysts. *Chem. Eur. J.* **2013**, *19*, 8030−8041. (b) Leboeuf, D.; Gandon, V. Carbon−carbon and carbon−heteroatom bond-forming transformations catalyzed by calcium(II) triflimide. *Synthesis* **2017**, *49*, 1500.

(16) Ohwada, T.; Okamoto, I.; Shudo, K.; Yamaguchi, K. Intrinsic pyramidal nitrogen of N-sulfonylamides. *Tetrahedron Lett.* **1998**, *39*, 7877−7880.